

INTEGRATING GENOMIC PREDICTION AND CROP GROWTH MODELS TO MAXIMIZE SOYBEAN SEED YIELD POTENTIAL

INTRODUCTION

Overview

Current rates of genetic improvement are potentially slower than required to ensure national food security (Bailey-Serres et al., 2019). The genetic improvement of complex traits like seed yield is a long-term process that requires multiple cycles of breeding and extensive field trials within a target environment (Vieira and Chen, 2021). Therefore, there is always interest in scientific advances to accelerate breeding and improve the efficiency of resource allocation (Cooper et al., 2021a). Limited understanding of trait genetic architecture and the physiological basis of yield hampers breeding programs (Hammer et al., 2019). While dynamic crop growth models lack a proper genetic framework, genomic-based prediction models lack functional sensitivities of physiological processes to environmental stresses such as temperature and water deficit (Boote et al., 2021). The integration of genomic prediction and crop growth modeling establishes a new frontier for breeders, crop modelers, and crop physiologists to collaboratively add value to breeding programs through precision breeding (Messina et al., 2018). In this proposal, we will leverage an integrated approach to accelerate the accurate identification and development of superior soybean genotypes, thereby supporting the sustainability and competitiveness of the U.S. soybean value chain.

Upon completion of this project, we will deliver: (1) quantitative data on the contribution of temperature and photoperiod to existing variability in soybean phenology traits, (2) a new genomic prediction-based crop model framework to enhance the connection between soybean physiological traits and their genetic regulation, and (3) soybean ideotypes for target environments based on our newly developed framework. The information generated will help breeders to quantify and understand the variation of germplasm responses across environments (Cooper et al., 2014). This enhanced understanding will help introduce genetic diversity and improve selection by narrowing down promising lines and anticipating potential impacts of trait variation in unobserved environments. We expect to advance understanding of the underlying traits driving soybean seed yield (Lucini et al., 2021), which is essential to sustain long-term soybean yield gains in the U.S.

Phenology development is the backbone of physiological processes driving emergent outcomes like leaf area development, light capture, and yield formation (Hammer et al., 2010). Understanding soybean phenology and its genetic control is an essential first step to understanding and increasing soybean seed yield potential. In crop simulation models, the most important parameters to set when modelling a new cultivar are those affecting crop development and phenology (Boote et al., 2013). From a crop physiology perspective, optimizing soybean developmental stages to maximize light capture and resource allocation efficiency to seed formation seems a reasonable strategy to increase yield (Boote et al., 2003), but environmental tradeoffs exist (Egli et al., 1984). Our knowledge of how the exploitable variation in phenology traits affects seed yield formation and how it interacts with a given environment is limited. The absence of this knowledge limits our ability to identify actionable strategies to accelerate precision breeding and current rates of genetic improvement. Thus, there is a critical need to develop a mechanistic genotype-to-phenotype system framework to identify solutions contributing to a consistent yield advantage in soybeans for target environments.

The originality of this proposal is grounded on the integration of crop growth models and genomic prediction to identify mechanisms and desirable traits to optimize seed yield formation. Our long-term goal is to help stakeholders in U.S. agriculture overcome yield-limiting challenges by better understanding the physiological and genetic mechanisms underlying seed yield across environments. This project was developed in close alignment with agricultural stakeholders. The Arkansas Department of Agriculture has expressed strong support for this effort, emphasizing the urgent need to develop novel integrative approaches to accelerate soybean improvement and maintain the competitiveness of U.S. soybean production (see support letter). Our overall goal is to identify strategies to increase soybean seed yield potential by integrating genomic prediction and crop growth models. Our central hypothesis is that a consistent soybean seed yield advantage can be achieved by optimizing phenology life-cycle traits over the crop season to maximize light capture and seed filling. We propose the following objectives:

Objective #1: Quantify temperature and photoperiod impacts on developmental rates of diverse soybean accessions. Working hypothesis – genotypic variation in photoperiodic sensitivity at post-flowering will drive differences for reproductive processes across the soybean accessions (Thomas and Raper, 1976).

Objective #2: Develop a genomic prediction-based crop model framework through the implementation of genomic prediction models into the APSIM-Soybean crop growth model. Working hypothesis – the genomic prediction-based crop model will enable prediction of soybean seed yield and associated traits comparable with the classic APSIM-Soybean model while advancing the explanatory power of genetic regulation for seed yield (Messina et al., 2006).

Objective #3: Identify soybean ideotypes for target environments using an integrated genomic prediction-based crop modeling framework. Working hypothesis – genotypes with longer seed fill duration will provide a consistent seed yield advantage in the U.S. Mid-South soybean production systems, with environmental tradeoffs modulating the critical maximum filling period for highest yield (Boote, 1981; Egli et al., 1984).

The fundamental rationale of the proposed project is that once our genomic prediction–crop modeling framework is created and tested, we can identify optimum phenology trait ranges maximizing seed yield under specific environmental conditions, providing opportunities to accelerate precision breeding and enhance soybean seed yield. Our combined expertise in soybean breeding, crop modeling, and physiology makes us the ideal team to develop this framework.

Review of Relevant Literature

The need for enhanced connection of complex traits to their genetic regulation

Breeders and crop modelers commonly direct attention to final seed yield, given that yield is the outcome that will generate economic returns. However, seed yield is among the most complex agronomic traits because it integrates the interactions of all eco-physiological processes and their genetic regulation. The individual contribution of quantitative trait loci (QTLs) for yield is low and largely dependent on the environment (Jarquín et al., 2014; Vieira et al., 2022a). Using crop growth models to dissect the physiological basis of complex traits could lead to the identification of putative causes that are not intuitively obvious from the perspective of maximizing seed yield (Hammer et al., 2010). We argue that understanding the genetic basis of shifts in soybean life cycle associated with phenology traits will explain emergent phenotypic differences in leaf area development, biomass production, and seed yield. Crop phenology traits generally have a

relatively simple inheritance (Pfeiffer and Egli, 1988) and can be measured precisely (Vallejos et al., 2022). Using a dataset with 39 cultivars and 14 trials, Hoogenboom et al. (1997) found that a gene-based model accounted for 75% of the variation in time to flower, 68% for maturity, 39% for seed size, and 11% for seed yield in common beans (*Phaseolus vulgaris* L.).

Previous studies have identified the role of *E* loci associated with soybean maturity and variation in vegetative and reproductive developmental stages (Curtis et al., 2000; Li et al., 2019; McBlain et al., 1987; Sedivy et al., 2017; Wang et al., 2019; Zimmer et al., 2021), some of which are associated with photoperiod sensitivity (Abe et al., 2003; Cober et al., 1996b, 1996a; Cober and Voldeng, 2001; Saindon et al., 1989). Zimmer et al. (2021) provided a comprehensive summary of 12 major loci associated with soybean maturity. Although most previous studies focused on flowering and maturity stages, some recent work has leveraged high-resolution phenology data to map loci influencing other developmental stages. For instance, Li et al. (2019) conducted a genome-wide association study (GWAS) with 278 accessions genotyped at 34,710 single-nucleotide polymorphisms (SNPs), and identified 37 significant SNP peaks associated with soybean flowering time and other developmental stages, including full flowering, beginning pod, beginning seed, and full seed.

Breeding to improve the balance between pre- and post-flowering phases to maximize soybean seed yield has demonstrated potential in previous studies (Boote, 2011; Kantolic and Slafer, 2001). Pfeiffer and Egli (1988) identified a 26-day range in the effective seed filling period across four soybean populations in the F₂, F_{2:3}, and F_{2:4} generations planted in Lexington, Kentucky. These authors identified significant heritability estimates for the R5 to R7 duration period in three of the four populations (0.16-0.63), and for all populations for the duration of the R1 to R8 period (0.37-0.82). Egli et al. (1984) identified variation of up to 33 days (23-56 days) in the R5-R7 period of 59 genotypes grown in Lexington, Kentucky. The authors identified a significant correlation (0.57) between seed yield and the seed filling period in 1 out of 3 experimental years. This suggests that selecting accessions with long seed filling periods could lead to higher-yielding soybean cultivars, but this is not consistent across all environments due to genotype × environment × management (G×E×M) interactions.

These literature results indicate the following: (i) variability exists in phenology traits in the soybean germplasm with moderate to high heritability for key developmental stages, (ii) optimizing phenology provides alternative pathways to maximize seed yield but G×E×M interactions, (iii) mapping efforts using detailed phenological data beyond flowering and maturity can uncover new regulatory regions for soybean phenology plasticity, but few studies have performed fine-scale mapping using detailed phenological data beyond flowering and maturity, and (iv) our knowledge of how altered phenology traits affect seed yield formation across environments is very limited. Integrating genomic prediction and crop modeling efforts presents a critical opportunity to fill this knowledge gap.

Given the interactive impacts of temperature and photoperiod on soybean development, quantifying the role of these factors is the essential first step to identifying genetic mechanisms controlling soybean development and seed filling dynamics. A substantial amount of effort has been dedicated to understanding the genetic controls of flowering and maturity time based on days after sowing (DAS) or emergence (DAE). However, DAS or DAE does not provide a stable phenotype because different genotypes respond to temperature and photoperiod in different ways. Of particular interest, we propose to use crop growth models to overcome this issue by generating genotype-specific coefficients that can be used as stable phenotypes, potentially giving more

power in genomic analyses (Wang et al., 2025). These genotype-specific coefficients are derived using a mechanistic approach to predict developmental stages based on photothermal accumulation (Archontoulis et al., 2014). This effort will greatly advance our understanding of the interactive effects of temperature and photoperiod on genetically diverse soybean accessions.

Studies integrating genomic-based predictions and crop modeling have been reported for multiple crops with varying levels of success (Bogard et al., 2014; Laurent et al., 2025; Messina et al., 2006; Oliveira et al., 2021; Vallejos et al., 2022; White and Hoogenboom, 1996). The first attempt to combine an ecophysiological soybean model with a gene-based model to predict cultivar-specific parameters was made by Messina et al. (2006). The study was conducted in Gainesville, Florida, using 14 near-isogenic lines planted in 2 years over 2 sowing dates. The authors combined the model CROPGRO-Soybean within the Decision Support System for Agrotechnology Transfer (DSSAT) framework with linear models that predict cultivar-specific parameters as functions of *E* loci. The CROPGRO-Soybean model predicted various developmental stages with a root mean squared error of approximately 5 days using the *E* loci-derived parameters. This study provided an important proof of concept, showing the potential of linking genetic information to a soybean crop model for predicting phenology. However, opportunities for advancement remain due to the small genetic panel, the limited number of environments, and the use of simple linear models for predicting cultivar-specific parameters.

Building on these early efforts, our proposal aims to advance knowledge on the following aspects: (a) implementing advances in the field of functional genomics and genomic prediction (Vieira et al., 2025, 2022a, 2022b) to estimate genotype-specific coefficients in crop growth models, (b) quantifying photothermal interactions using an unexplored genetic pool, (c) developing a comprehensive benchmark analysis of the genomic prediction-based crop modeling framework with other well-established genomic prediction models for seed yield, and (d) using a modern plant modeling framework (Brown et al., 2014) available in the latest APSIM version (APSIM Next Generation, Holzworth et al., 2018) to implement genomic prediction models, which brings substantial improvements in execution speed and script flexibility for model adaptation.

The need for an integrative genomic prediction–crop modeling framework

The Agricultural Production Systems sIMulator (APSIM) is an advanced cropping systems modeling framework (Holzworth et al., 2014) that considers weather, soil, management, and crop inputs for simulating soil water balance, surface residue decomposition dynamics, soil carbon, nitrogen cycling, crop evapotranspiration, and the growth and development of various crops (Figueiredo Moura da Silva et al., 2025; Keating et al., 2003; Kothari et al., 2025; Probert et al., 1998). The APSIM-Soybean model calculates biomass production based on intercepted daily solar radiation, radiation use efficiency (RUE), and the impact of environmental stressors like water deficit, temperature, and nitrogen (Holzworth et al., 2014). This model has proven successful in predicting soybean growth, development, and seed yield across various environments in the U.S. (Archontoulis et al., 2020; Elli et al., 2022). Crop growth models commonly present a sufficient representation of physiological processes, enabling the prediction of complex traits as an emergent consequence of underlying fundamental processes (Hammer et al., 2010). While these models have been used for decades to predict the implications of genotype selection and crop management on soybean yield (Boote et al., 1996; Elli et al., 2022), their use in plant breeding for genotype-to-phenotype prediction is in early stages. Our goal is to develop a novel framework that integrates genomic prediction-based information with an ecophysiology-crop model to predict soybean

phenotypes under various environments to facilitate the conceptualization of strategies to accelerate the accurate identification and selection of superior genotypes.

A bottleneck in using crop growth models for predicting crop yield is that their testing process requires coefficients that are genotype-specific (Jighly et al., 2023) and require extensive and expensive experimentation to stay up-to-date with new genotypes developed annually (Messina et al., 2006). A fundamental conclusion from the calibration process in crop models is that the genotype-specific parameters hold genetic information. However, because of the phenotypic nature of the crop model parameters, it is unknown what processes they represent at the molecular level and what causes their variation, limiting the applicability of existing crop models in plant breeding. A quantitative genetics analysis would reveal the genomic regions that control the genotype-specific parameter values within crop models. This approach will improve connection to the genetic architecture that controls the trait of interest by identifying model parameters that link more effectively to genomic regions than direct phenotypic measures. For example, the generated genotype-specific model parameters would reveal the contribution of temperature and photoperiod to soybean developmental rates, which are more stable traits than the phenotypic measures of time to flowering or physiological maturity (Boote et al., 2021).

While genomic prediction has emerged as a revolutionary tool to assist plant breeders in predicting phenotypes from untested genotypes, thus accelerating genetic gain, challenges remain (Cooper et al., 2021b). Economically important traits are commonly controlled by many genomic regions, where the environment often modulates changes in magnitude and direction (Xavier et al., 2018). One consequence of not having a framework integrating trait physiology in genomic prediction models is the potential misinterpretation of trait plasticity under stress conditions (Hammer et al., 2019). Xavier et al. (2018) and Mir et al. (2012) highlighted the need for transdisciplinary investigations and the inclusion of management and environmental variables to better predict yield stability and stress tolerance in soybeans. The narrow genetic base of modern soybean genotypes also exacerbates the challenge of identifying genomic regions associated with traits for environmental stresses, particularly abiotic stresses (Manavalan et al., 2009).

Preliminary Results

A field study was conducted in 2024 using a panel of 250 genetically diverse soybean accessions from maturity groups III and IV in two locations in Arkansas (Fayetteville and Stuttgart), covering a geographical gradient of approximately 380 km. The accessions were laid out in a randomized complete block design with two replications. At harvest maturity (R8 stage), total aboveground biomass was collected from approximately 1 m² of plot area. Samples were dried at 50°C for 7 days, weighed, and threshed. Subsequently, seed mass was determined (Edwards and Purcell, 2005). Harvest index (HI) was calculated as the ratio of seed mass to total aboveground biomass. For seed yield determination, plots were harvested using a small-plot combine, and seed weights were corrected for 130 g kg⁻¹ moisture content. A wide phenotypic variation in the selected accessions was observed (Figure 1). Our results indicated that some

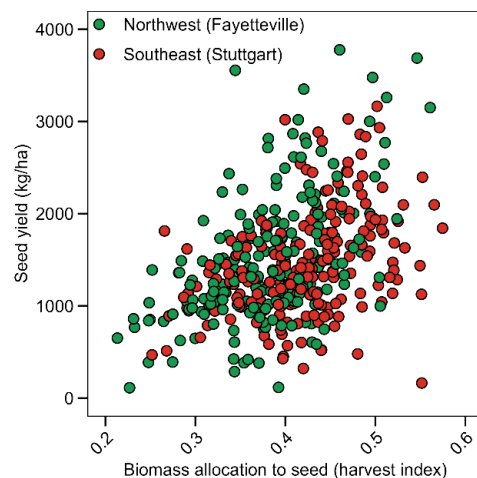


Figure 1. Phenotypic variability in seed yield and harvest index across the 215 genetically diverse soybean accessions evaluated.

accessions can allocate assimilates to seed more efficiently than others. Consistent with genetic gain studies that show increases in HI with increases in seed yield (Rowntree et al., 2014; Umburanas et al., 2022), we identified a positive association between HI and seed yield, suggesting the potential of this trait to maximize seed yields. Compiling these preliminary results and our literature review, we argue that seed fill duration will help explain variability in HI and thus final seed yield (Curtis et al., 2000; Tamagno et al., 2018). Therefore, an enhanced understanding of trait genetic architecture controlling seed fill will facilitate conceptualization of strategies to maximize yields.

The University of Arkansas soybean breeding group, led by Co-PI Vieira, develops around 200 bi-parental populations each year, which results in roughly 1,000 preliminary breeding lines entering multi-environment yield trials and molecular marker characterization. Over the past three years, the program has developed a comprehensive training set consisting of over 3,500 breeding lines, including genotypic and multi-environment phenotypic data. The team has done extensive work using machine learning and statistical learning-based genomic predictions for various traits, including grain yield and stress resilience, utilizing Random Forest and Support Vector Machine algorithms (Vieira et al., 2025, 2022a, 2022b). Thus, we are equipped with the resources (phenotypic and genotypic data) and knowledge to support this proposal.

We ran an in-silico analysis in Central-East Arkansas with the APSIM-Soybean model to further explore the hypothesis that shifts in soybean phenology and seed fill duration can maximize seed yield (Egli et al., 1984). Our simulation supports our central hypothesis that soybean seed yield advantage can be achieved by optimizing life cycle phenological traits, but it also identified a key interaction with soil water supply (**Figure 2**). We designed hypothetical genotypes with photothermal (PT) time required to complete seed filling (R5 to R7) ranging from 350 PT to 850 PT degrees (**Figure 2**). For reference, a maturity group VI genotype common value for seed fill period is ~540 PT degrees (Elli et al., 2022). We investigated two scenarios: (1) altering the seed filling period while the normal time to flowering was maintained, referred to as “Life cycle extended,” and (2) altering the seed filling period while time to flowering was reduced to keep normal overall crop cycle duration, here referred to as “life cycle fixed.” The rationale behind the second scenario is that in northern environments of the US, the extended life cycle could prevent soybean plants from completing their life cycle before the occurrence of killing frosts in the fall (Boote et al., 2003).

An advantage U.S. Mid-South farmers have over farmers in many other areas of the U.S. is a longer frost-free growing season that provides abundant solar

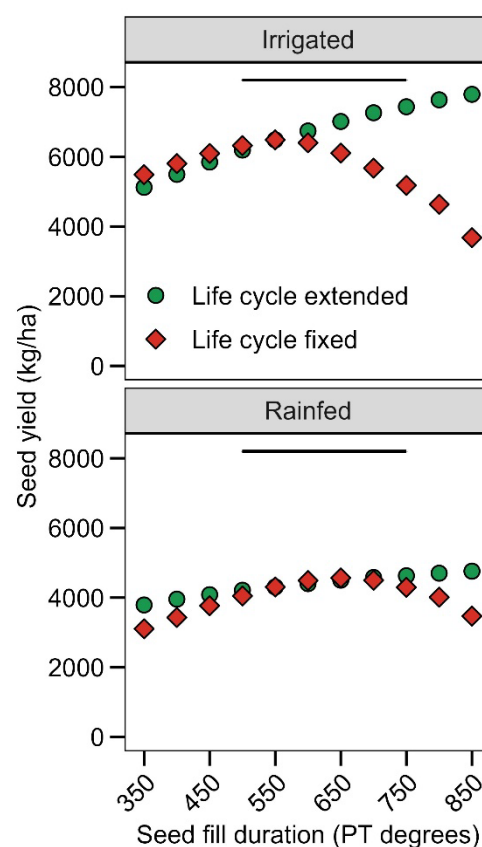


Figure 2. Soybean seed yield response to variation in seed fill duration in photothermal (PT) degrees in Central-East Arkansas. Horizontal line indicates the range of genetic variability from Wu et al. (2019). Symbols indicate average values across 40 weather years (1985-2024).

radiation, facilitating an extended soybean life cycle. Our results indicate that increasing seed fill (**Figure 2**) increased yield, especially in well-watered environments. When the seed filling period was increased with a concurrent reduction in the time to flowering, optimum yield was obtained at around 550 and 650 PT degrees for irrigated and rainfed treatments, respectively, with subsequent decreases particularly in well-watered conditions. The main reason was a trade-off between the earlier start of seed filling and detrimental impacts on leaf area development during vegetative growth. Reduced leaf area index (LAI) became a limiting factor to light capture during seed filling. The less pronounced impact in water-limited environments was because lower LAI formation combined with early flowering provided an opportunity to save soil moisture used during critical periods like seed filling. Extending the post-flowering life cycle enhanced persistence of green leaf area, which increased crop growth rate during the late seed filling period without detrimental impacts on vegetative leaf area development (**Figure 3**).

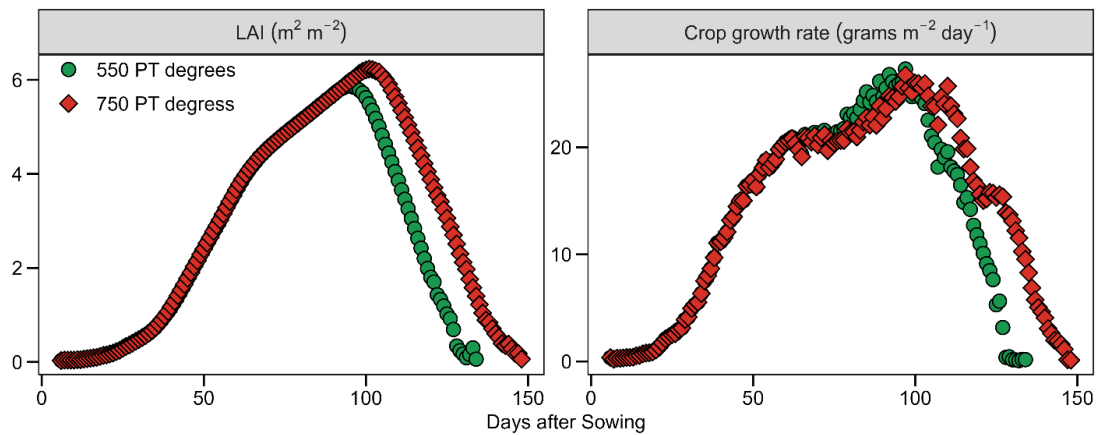


Figure 3. Daily dynamics in leaf area index (LAI, left panel) and crop growth rate in grams of aboveground biomass $\text{m}^{-2} \text{day}^{-1}$ (right panel) using a hypothetical cultivar with 750 photothermal (PT) degrees instead of 550 PT degrees, while the normal time to flowering was maintained (approximately 12-day increase in the whole-life cycle). Symbols are the average across 40 years (1985-2024) of simulations in Pine Tree, Central-East Arkansas, under irrigated conditions.

Recent advances in high-throughput phenotyping using UAV imaging and proximal sensing can facilitate characterization of canopy development at a fine temporal scale (days or weeks), providing new opportunities to identify loci controlling dynamic developmental traits (Fritsche-Neto et al., 2025; Kaler et al., 2020; Li et al., 2023). Co-PI Fritschi has performed extensive work integrating Satellite/UAV Data Fusion and Machine Learning approaches to monitor crop development and soybean leaf area index - LAI (Hoyos-Villegas et al., 2014; Maimaitijiang et al., 2020). Our proposal aims to leverage high-dimensional genomics, phenomics, and environmental data within crop growth models, which could lead to improved accuracy in yield predictions and accelerate genetic gains (Fritsche-Neto et al., 2025; Hammer et al., 2019; Messina et al., 2018).

We propose to use the APSIM NG-Soybean model in this project for the following reasons: a) the model has proven robust in simulating several soybean traits across a range of soybean production systems in the U.S., including soil and crop N dynamics, soil water dynamics, leaf area index, phenology, aboveground biomass, and seed yield (Archontoulis et al., 2020; Elli et al., 2022; Freitas et al., 2025; Pasley et al., 2021), b) the newest model version has modern languages and frameworks that allow for much greater script flexibility providing an ideal environment for extending model capabilities to different levels of complexity (Brown et al., 2014); c) the model

is used in more than 120 publications per year, it receives more than 3,500 citations per year, and its use in research is exponentially growing based on Web of Science metrics; and d) PI Elli has more than 6 years of experience in calibrating, improving, and applying the APSIM framework (Elli et al., 2020a; Elli et al., 2022, 2019; Elli and Archontoulis, 2023).

Our preliminary results demonstrate: 1) the value of using a crop growth model to advance knowledge of fundamental plant traits affecting yield formation, 2) the need to create frameworks that integrate genetics and crop physiology to improve genotype-to-phenotype predictions and conceptualize more efficient solutions for accelerating breeding, and 3) the need to better quantify genotype by environment tradeoffs to better understand the impact of stressors like water deficit (e.g., **Figure 2**) to identify traits for target environments. These findings provide the rationale for our proposed work to integrate genomic prediction and crop modeling and identify traits that can enhance soybean yield potential under diverse environments.

RATIONALE AND SIGNIFICANCE

Rationale

Our rationale is that improved understanding of how phenology traits influence soybean seed yield formation across target environments will enable breeders to better quantify and understand germplasm responses across environments. This new knowledge will facilitate the introduction of greater genetic diversity and improve selection efficiency by identifying promising genotypes and anticipating the effects of trait variation in new environments, ultimately accelerating precision breeding and enhancing soybean yield potential.

Our proposal addresses the program “**Plant Breeding for Agricultural Production**” by supporting breeding efforts to improve crop productivity, efficiency, and performance. We address this by investigating traits to increase soybean seed yield potential for target environments. Our proposal has the **potential to further support or improve rates of genetic gain** through integrating genomic prediction and crop growth models. Our proposal addresses **the key requirements** including: (1) utilizing useful genetic diversity, (2) improving selection accuracy and intensity through direct or indirect phenotyping across relevant time scales and environments, (3) identifying, modeling, and exploiting, heritable epigenetic and genetic marks associated with desirable traits, and (4) identifying, modeling, and implementing breeding methods that have potential to reduce the effective generation interval or timeline for cultivar release. We address the priority area of **leveraging plant genetic resources** by investigating diverse soybean accessions with known genetic information to better understand the genetic control of phenology and incorporate this knowledge into predictive modeling frameworks to improve selection for high-yielding genotypes.

Significance

Despite the growing interest in opportunities to accelerate breeding and improve complex traits like seed yield, our limited understanding of the physiological basis underlying yield formation hampers breeding programs (Hammer et al., 2019). Combining genomic information with crop growth models to enhance understanding of trait genetic architecture and the physiological basis of trait contributions to yield formation can accelerate breeding by enhancing quantification of genotype by environment interactions. *Our contribution is significant because new knowledge will directly inform initiatives and recommendations for breeders to better understand soybean seed*

yield response in their germplasm across environments to accelerate precision breeding. In the short term, our integrative genomic prediction–crop modeling framework will generate information that breeders can adopt to facilitate the introduction of genetic diversity and improve selection by anticipating germplasm response in target environments and narrowing down promising lines. In the long term, we expect our framework to improve soybean seed yield and resilience by identifying optimum trait ranges and facilitating conceptualization of plant ideotypes for target environments.

APPROACH

The proposed approach involves three major steps: 1) conduct field experiments to collect phenology information and seasonal canopy characteristics from a diverse soybean panel to identify photothermal interactions across exploitable genetic variation for key soybean phenology stages, 2) developing a genomic prediction-based crop model framework, and 3) using an integrated genomic prediction-based framework to inform breeding by establishing benchmark analysis and identifying ideotypes for target environments (**Figure 4**).

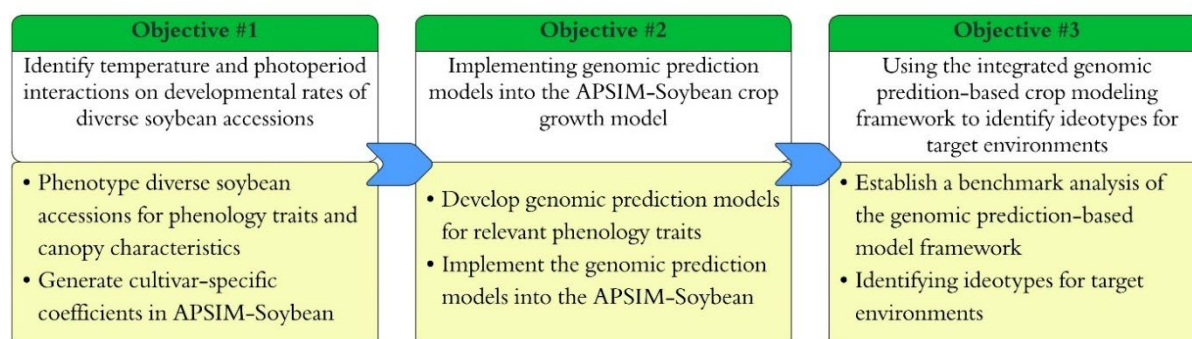


Figure 4. Objectives and main deliverables for developing and applying an integrated genomic prediction-based crop modeling framework for soybean.

Objective #1: Identify temperature and photoperiod interactions on developmental rates of diverse soybean accessions

Introduction. Previous research has shown important but limited information on photothermal interactions modulating the possible extent of exploitable genetic variation in soybean phenology (Messina et al., 2006). Our goal here is to dissect the contribution of temperature and photoperiod on existing variability in soybean phenology traits by evaluating a comprehensive set of soybean accessions. Our working hypothesis is that genotypic variation in photoperiodic sensitivity at post-flowering will drive differences in developmental rates. To test our working hypothesis, we will conduct field experiments with various genetically diverse soybean accessions across contrasting temperature and photoperiod conditions and phenotype them for key phenology stages, canopy characteristics, and yield (*task 1.1*). A novel outcome here is the generation of new APSIM phenology coefficients for the evaluated soybean accessions (*task 1.2*). New knowledge will lead to the identification of existing variability in the life cycle dynamics as affected by photothermal interactions. The absence of this knowledge limits our ability to identify potential plant ideotypes to maximize soybean yield potential. Upon completion of this objective, we will generate a database of exploitable variability in developmental stages and calibrated phenology coefficients of the APSIM-Soybean model. This result is expected to make a positive impact because the new knowledge can be used to identify avenues to maximize soybean seed yield potential.

Task 1.1: Phenotype diverse soybean accessions for phenology traits. In years 1 and 2 of the project, we will grow 250 genetically diverse soybean accessions and elite cultivars from the USDA Soybean Germplasm Collection at two field sites, one in Northwest Arkansas (Fayetteville, where PI Elli and Co-PI Vieira are housed) and one in Columbia, Missouri (Co-PI Fritschi). The rationale for selecting these accessions is grounded in the following reasons. First, the selected panel has a broad coverage of phenotypic diversity within soybean maturity groups 3 and 4 (**Figure 1**), and molecular marker information is available for the selected genotypes. Second, we have enough seed already available for all genotypes to conduct the proposed field experiment, as a result of seed increase over the last 2 years from a project funded by the Arkansas Soybean Promotion Board, in which Co-PI Vieira and PI Elli are collaborating. This will allow the project to be completed within the proposed timeline. Lastly, substantial variability in developmental stages was previously obtained for a smaller set of genotypes (Cober et al., 1996a; Curtis et al., 2000), serving as the foundation needed to expand phenology characterization across a broad range of genotypes and environments.

Two planting dates will be evaluated to exploit differences in photoperiod during the crop season: early, at around April 15th, and late, at around June 15th. The selected sites and sowing dates will span a 23.5 to 28.4 °C average season temperature and photoperiod variation from 10.7 to 14.9 hours. The experiments will be conducted within the university's research stations network. We will target a plant density of 30 plants m⁻², and the field experiments will be conducted under irrigated conditions. Irrigation will be managed following recommendations adopted by each research station to avoid confounding impacts of water limitation on crop growth and development. The experiment will have two factors, soybean accessions and planting dates, laid out in a randomized complete block design with two replications. We will have 250 accessions * 2 planting dates * 2 replications = 1000 plots per location per year. Individual plots will have 4 rows of 6 m and a row space of 76 cm.

The following developmental stages will be recorded using the Fehr and Caviness (Fehr and Caviness, 1977) scale: emergence (50% of plants with cotyledons above the soil surface), beginning flowering (R1, first open flower on the main stem), beginning pod stage (R3, pods of at least 5 mm in any of the four uppermost nodes), beginning seed stage (R5, seeds of at least 3 mm in any of the four uppermost nodes), and physiological maturity or R7 (one mature pod in any node on the main stem in 50%). The developmental stages will be recorded at the whole canopy level (Salmerón and Purcell, 2016) 3 times a week, and all stages will be considered reached when ≥ 50% of the plants in the plot meet each stage's criteria. The occurrence dates for each phenological stage will be converted to days after emergence. Because of the high variability between planting and emergence date, we will start counting days after the emergence date rather than the planting date (dos Santos et al., 2019). From emergence to physiological maturity, canopy characteristics will be measured based on weekly drone flights. Multi-sensor drone platforms with integrated RGB and IR (FLIR SIRAS) sensors will be flown in Arkansas, whereas RGB and IR (FLIR DUO) and multispectral (Micasense Altum) sensors will be flown in Missouri. Ground and color reference panels will be placed in the field, and orthomosaics will be generated to determine plot-level canopy coverage as well as greenness, as described in previous work done by Co-PI Fritschi (Hoyos-Villegas et al., 2014; Maimaitijiang et al., 2020). Non-destructive LAI will be derived from drone imagery by combining spectral information with data fusion and machine learning techniques, similar to the approach used by Maimaitijiang et al. (2020). Final seed yield will be determined at harvest maturity (R8) using a small-plot combine, and seed weights will be corrected to 13% moisture.

Task 1.2: Generate cultivar-specific coefficients in APSIM-Soybean. We will use phenology information collected in **task 1.1** to generate cultivar-specific coefficients in APSIM-Soybean. APSIM-Soybean model uses a multiplicative function to describe the daily relative development rate (R) as a function of temperature, $f(T)$, and photoperiod, $f(PP)$:

$$R = f(T) \times f(PP) \quad (1)$$

This method uses daily temperature and photoperiod to calculate a daily rate of development, which is accumulated each day from emergence. The timing of key development phases, such as flowering and physiological maturity, is predicted by integrating R over time. Temperature and photoperiod functions currently implemented into the APSIM-Soybean model are presented in **Figures 5A and 5B**, respectively. Developmental rates are a function of temperature and photoperiod, expressed as photothermal time. A value of 1 represents maximum development rate, while 0 represents null development (**Figure 5**). APSIM-Soybean has a base temperature of 10°C , an optimum temperature for development of 30°C , and a maximum temperature for development of 40°C . A 3-h temperature function (dos Santos et al., 2022) using cubic interpolation from minimum and maximum daily temperatures is used by APSIM-Soybean to account for situations when the temperature moves across cardinal values of $f(T)$ during the day (e.g., minimum temperature $< T_b$).

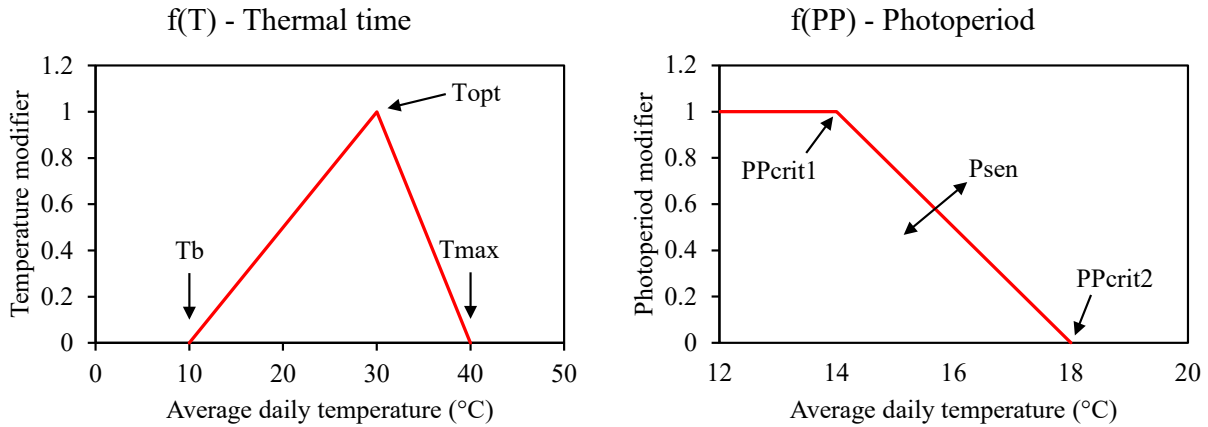


Figure 5. Temperature (A) and photoperiod (B) functions used by APSIM-Soybean to simulate life-cycle developmental stages from sowing to maturity.

APSIM-Soybean model simulates phenology from sowing to harvest based on 15 key parameters (**Table 1**). The APSIM-Soybean version we will use (APSIM-NG) accounts for simplified phenology coefficients by taking out stages that are not measurable, and by adding new stages that are measurable, following a staging system (Fehr and Caviness, 1977). The model requires cultivar-specific calibration phenology parameters that represent the photothermal time required for the completion of two consecutive phenology stages (**Table 1**). A development stage is completed after the required photothermal time is reached to signal movement to the next stage. The photoperiod modifier function (**Figure 5B** and **Table 1**) is controlled by 3 parameters (P_{crit1} , P_{crit2} , P_{sen}) that set cultivar sensitivity to photoperiod and can be used to characterize genotypes in terms of photoperiod response and its interaction with temperature. Soybean is a short-day plant, meaning shorter days accelerate development rates towards flowering and physiological maturity. APSIM-Soybean requires calibration for photothermal time ($^{\circ}\text{C}\cdot\text{day}$) from emergence to start flowering, from start flowering to start pod development, from start pod development to start grain

fill, and from start grain fill to physiological maturity, which can be controlled by single or multiple parameters (**Table 1**).

The occurrence dates, converted to days after emergence, for the phenological stages phenotyped in *task 1.1* will be used to derive the cultivar-specific model parameters (**Table 1**) using an iterative optimization procedure based on the Nelder–Mead simplex algorithm (Cui et al., 2023). The optimization procedure aims to minimize the residual sums of squares between estimated and actual timing to reach a given stage for each soybean accession. We will run the algorithm using R software (R Development Core Team, 2021) and the APSIMX package (Miguez, 2020). Daily weather data required as input for APSIM (maximum and minimum temperature, solar radiation, and rainfall) will be obtained from local weather stations.

Table 1. APSIM-Soybean model phenology parameters used by APSIM-Soybean to simulate life-cycle developmental stages from sowing to maturity.

Category	Parameters	Definition	Unit	Typical value or range
Temperature	Tb	Base temperature below which the development rate is zero	°C	10
	Topt	Optimum temperature for development	°C	30
	Tmax	Maximum temperature above which the development rate is zero	°C	40
Photoperiod	Pcrit1	Photoperiod above which the development rate is sensitive	hours	14.7-22.4
	Pcrit2	Photoperiod above which the development rate is null	hours	11.8-14.6
	Psen	Photoperiod sensitivity coefficient	1/hour	0.13-0.35
Phenology	Emergence	Time from germination to emergence	°C-day	30-60
	Vegetative	Time from emergence to start flowering (R1)	°C-day	310-470
	EarlyFlower	Time from start flowering to start pod development (R3)	°C-day	100-200
	EarlyPodDev	Time from start pod development to start grain filling (R5)	°C-day	100-150
	EarlyGrainFill	Time from start pod development to end canopy development	°C-day	295-374
	MidGrainFill	Time from end canopy development to end pod development	°C-day	148-187

LateGrainFill	Time from end canopy development to end grain fill	°C-day	148-187
EntireGrainfill	Time from start grain fill to end seed fill	°C-day	590-748
Maturing	Time from end grain fill to physiological maturity (R7)	°C-day	0-45

There is potential to optimize 15 parameters (**Table 1**). Our initial focus will be on optimizing phenology- and photoperiod-related parameters, while keeping cardinal temperature-related parameters constant, giving little evidence of changes across genotypes (Archontoulis et al., 2014). Parameter ranges will be set based on default generic cultivar values present in APSIM-Soybean (**Table 1**). The optimization workflow will follow an approach similar to Salmerón and Purcell (2016). Briefly, we will optimize parameters for predicting the time to begin flower first. Thereafter, the calibrated coefficients affecting the prediction of begin flower will be fixed before setting the optimization to calibrate for subsequent reproductive stages (R3, R5, and R7). Seasonal canopy coverage and remote-sensed LAI data (*task 1.1*) will be included in the model optimization process to identify whether prediction improvements are realized (Gasó et al., 2023, 2021). **Figure 3** in the preliminary results section demonstrates the seasonal (daily) LAI outputs from APSIM-soybean. If accuracy is improved, this can be used as a potential high-throughput product to optimize the crop model calibration process.

Expected results from objective 1. At the end of year 2, we will have data on 4 key developmental stages (R1, R3, R5, and R7) for all accessions, planting dates, locations, and years. This dataset will enable a robust characterization of the exploitable genetic variation for key soybean phenology stages and their associated response to temperature and photoperiod. We expect to identify variability in developmental stages across the diverse soybean accessions, given that variability was previously obtained for a smaller set of genotypes (Cober et al., 1996a; Curtis et al., 2000). We expect to generate APSIM genotype-specific coefficients for each accession with sufficient precision (Root Mean Square Error less than 5 days for R1 predictions and less than 8 days for R7 predictions, Salmerón and Purcell, 2016). The generated coefficients will allow us to dissect the contributions of photoperiod and temperature-related coefficients for each accession.

Potential Pitfalls and Alternative Strategies. Unanticipated weather events (e.g., storms, hail) may preclude detailed phenotypic data collection from field experiments or delay planting. Having the experiment at two contrasting locations, two sowing dates, and two years will provide resilience to these factors and the completion of the proposed tasks. While phenology is a time-consuming measurement, the proximity of the experiment to where PI/Co-PIs are housed will ensure the accessions can be realistically phenotyped. Given findings reported in previous studies (Cober et al., 1996a; Curtis et al., 2000) and our strategic approach in composing the diversity panel, variation in phenology is ensured. A drawback of using optimization algorithms is the high computational cost of generating genotype-specific parameters. The Arkansas High-Performance Computing Center provides high-performance computing hardware to enable computationally intensive research. Parallelization options will be explored by integrating APSIM-Soybean with R using the package APSIMX (Miguez, 2020).

Objective #2: Develop a genomic prediction-based crop model framework through the implementation of genomic prediction models into the APSIM-Soybean crop growth model

Introduction. An integrative framework linking genomic prediction and crop growth modeling to support domestic soybean breeding is missing. Our goal here is to implement newly developed genomic prediction models for relevant phenology traits into the APSIM-soybean crop growth model. Our working hypothesis is that the genomic prediction-based crop model enables the prediction of soybean seed yield and associated traits while enhancing the explanatory power of soybean yield formation. We will test our hypothesis through two major tasks: 1) develop genomic prediction models to predict cultivar-specific coefficients in APSIM-soybean, and 2) implement the predicted values into the APSIM-Soybean model. The new knowledge obtained here will lead to an enhanced connection between soybean complex traits and their genetic regulation, which can facilitate the conceptualization of strategies to accelerate breeding. The proposed framework is needed because, without it, we will have limited ability to identify which traits fundamentally control yield formation in soybeans. Upon completion of this objective, we expect to have a genomic prediction-based crop modeling framework that simulates soybean growth and yield using the genetic makeup of the genotypes. This outcome is expected to add value to the field of agriculture and soybean breeding because it advances current methods and initiatives to accelerate soybean breeding.

Task 2.1: Develop genomic prediction models for relevant cultivar-specific phenology coefficients used in APSIM. After obtaining cultivar-specific parameters for APSIM-Soybean (task 1.2), we will develop genomic-based prediction models for each parameter. Genomic prediction models will be fitted using cultivar-specific coefficients from the two locations and two sowing dates from experimental year 1, while year 2 will be used as independent data for model validation. We will use a flexible modeling framework described in a previous study led by Co-PI Vieira (Vieira et al., 2022a). Briefly, a flexible framework that allows the inclusion of the main effect of molecular markers via covariance structures will be implemented. In addition, this framework enables the inclusion of additional covariates (i.e., environmental data) and their respective interactions. Suppose that the genomic effect g_i of the i^{th} line can be characterized by a linear combination of p molecular markers x_m (where $m = 1, 2, \dots, p$) and their corresponding effects β_m such that $g_i = \sum_{m=1}^p x_m \beta_m$, with $\beta_m \sim N(0, \sigma^2_\beta)$. If we include all the genomic effects into a single vector, we have $g = X\beta$. From results of the multivariate normal distribution, the vector of genomic effects $g = \{g_i\} \sim N(0, G\sigma^2_g)$, with $G = XX^t$ as the corresponding variance-covariance matrix. Thus, the linear predictor can be expressed as:

$$p_{ij} = \mu + E_j + L_i + g_i + gE_{ij} + \varepsilon_{ij} \quad (2)$$

where p_{ij} is the APSIM cultivar-specific parameter response of the i^{th} genotype in the j^{th} environment. This is modeled as the sum of a mean effect μ common to all genotypes, a random line effect $L_i \sim N(0, \sigma_L^2)$, a random environmental effect $E_j \sim N(0, \sigma_E^2)$, a genomic $\times$ environment interaction term gE_{ij} , and a residual error $\varepsilon_{ij} \sim N(0, \sigma_\varepsilon^2)$. This flexible framework allows the inclusion of additional covariates and their interactions through covariance structures, following Vieira et al. (2022a).

Task 2.2: Implementing the genomic prediction models into the APSIM-Soybean model.

We will use the plant modeling framework (Brown et al., 2014) available in the latest APSIM version (APSIM-NG) to implement a new genomic prediction-based module (Figure 6). We will use an approach similar to Oliveira et al. (2021). Briefly, the module will be developed to allow for the input of genomic prediction models for each soybean accession (Figure 6) to estimate the phenology-related parameters within the APSIM-Soybean model (Table 1). Subsequently, the crop growth model simulates phenotypes like days to flowering, days to maturity, and seed yield using weather (daily maximum and minimum temperature, solar radiation, and rainfall), management (e.g., sowing date, plant population), and soil (e.g., drained upper limit, wilting capacity, organic carbon, depth). This new module will be developed within the R environment (R Development Core Team, 2021) and linked into APSIM-Soybean using the APSIMX package (Miguez, 2020). Minor structural changes to the phenology module in APSIM-Soybean will be needed to replace the genotype-specific parameters with the generated genomic prediction models from *task 2.1*, which replaces a constant function with an expression function already pre-defined within the APSIM plant modeling framework to allow for the insertion of external equations. The approach will be similar to that used by PI Elli to add new algorithms and structures into other crop models within APSIM framework (Elli et al., 2020a; Elli et al., 2020b).

For model evaluation, we will compare the measured developmental stages (e.g., days to flowering, days to maturity, *task 1.1*) with the simulated values using the classic APSIM-Soybean and the traditional approach of calibrating cultivar coefficients directly using phenotypic data, and the new genomic prediction-based APSIM-Soybean, hereafter referred to as APSIM-Soybean-PG, which uses cultivar coefficients from genomic prediction models, similar to what was performed for common beans by Vallejos et al. (2022). The model's performance will be evaluated using the Coefficient of determination – R^2 , Willmott Agreement Index (Willmott et al., 1985), Modelling Efficiency Index – E (Nash and Sutcliffe, 1970), and Root Mean Square Error – RMSE.

Expected results from objective 2. Upon completion of this objective, we expect to generate the APSIM-Soybean-GP, a genomic prediction-based version of the APSIM-Soybean crop growth model. We expect the model performance of the APSIM-Soybean-GP to predict phenology to remain comparable to the predictions from the classic model (Vallejos et al., 2022) but with an enhanced connection to the soybean genetic architecture. We expect to generate genomic prediction models for relevant phenology traits while advancing the explored genetic pool and

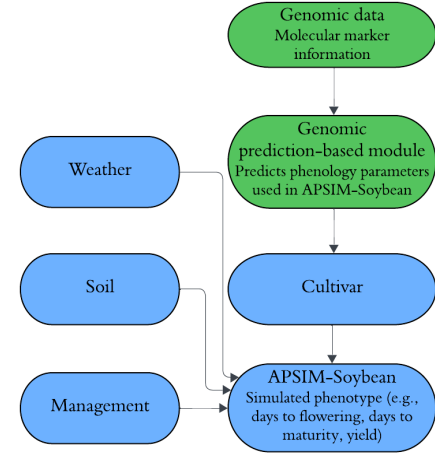


Figure 6. Overview of the new gene-based APSIM-Soybean model. Processes highlighted in green indicate the new modules added to the traditional APSIM-soybean model (blue).

applications of functional genomics and genomic prediction. We expect to identify genomic regions and markers associated with phenological stages and canopy characteristics, building on previously characterized loci and other SNPs identified through GWAS.

Potential Pitfalls and Alternative Strategies. Genomic prediction models may not exhibit the expected predictive ability for phenology traits, even though previous studies indicate that phenology can be predicted with sufficient accuracy (Vallejos et al., 2022, Hoogenboom et al., 1997). If this happens, we will explore the implementation of machine learning-based algorithms in combination with feature selection. As described by Co-PI Vieira (Vieira et al., 2022b, 2025), this approach has increased accuracy compared to a baseline GBLUP (genomic best linear unbiased predictor). Similarly, the APSIM-Soybean-GP model may not outperform the classic APSIM-Soybean for predicting phenology. A comparable performance to the classic crop growth model would add significant biological insight to simulate phenotypic consequences of genotype-by-environment-by-management interactions (Hammer et al., 2019). Also, the implementation of the framework sets the stage for similar approaches for other phenotypes and expands model capabilities that can be leveraged for other applications and environments.

Objective #3: Identify soybean ideotypes for target environments using an integrated genomic prediction-based crop modeling framework

Introduction. Systems approaches to design soybean ideotypes for target environments are of urgent need (Wu, 2023). Our goal is to use a well-tested genomic prediction-based crop modeling framework to inform breeding by identifying ideotypes for yield potential based on existing variability in soybean life-cycle developmental stages. Our focus is to design soybean plants that offer the most adequate trade-off in maximizing light capture by achieving a) adequate leaf area development during vegetative growth and b) adequate seed yield formation during reproductive growth for a given environment. Our hypothesis is that longer seed fill duration will provide a consistent seed yield advantage in US Mid-South soybean production systems. Our approach here will further test our genomic prediction-based model using observed yield data from a set of untested soybean genotypes across different environments, and it will develop a benchmark analysis with other well-established genomic prediction models (*task 3.1*). Subsequently, we will use our framework to better understand how shifts in phenology will affect emergent outcomes like leaf area development, canopy coverage, crop growth rates, and final seed yield towards identifying ideotypes for target environments (*task 3.2*).

Task 3.1: Establish a benchmark analysis of the genomic prediction-based crop modeling framework. We will use the integrated genomic prediction-based crop model to predict the seed yield of an additional set of 3,000 soybean breeding lines (see soybean breeding program capabilities described in *preliminary results*) that are genetically unrelated to the training set described in *Task 2.1*. The breeding lines are well-adapted to Mid-South conditions, and molecular marker and seed yield information are available for all the lines. A benchmark analysis will be performed against well-established genomic prediction models, including GBLUP - fitted with REML (restricted maximum likelihood) variance components (Habier et al., 2007), BayesA - Bayesian shrinkage regression (Meuwissen et al., 2001), BayesB - an extension of BayesA with variable selection (Meuwissen et al., 2001), and RF - random forest regression (Breiman, 2001). We will fit the models using the *BGLR* (Pérez and De Los Campos, 2014) and *ranger* (Wright and Ziegler, 2017) packages. We will then compare the predicted yield from the genomic prediction models with the yield simulated from the integrated genomic prediction-based crop model. For model evaluation, we will follow an approach similar to Xavier et al. (2016). Briefly, the predictive

ability will be evaluated as the correlation between observed and predicted values computed through cross-validation. We will conduct a 10-fold cross-validation with 5 replications.

Task 3.2: Identifying ideotypes for target environments. We will run an optimization to identify phenology trait values in the APSIM-Soybean crop model that maximize seed yield for target environments. We will use a similar approach to the optimization performed to calibrate cultivar coefficients in *task 1.2*. Briefly, we will use the Simplex Optimization Algorithm to minimize the residual sums of squares to identify the set of parameter combinations that maximize seed yield per environment. The optimized trait values will then be linked to genomic information using the genomic prediction models developed in *task 2.1*, allowing us to identify molecular markers associated with optimum APSIM-Soybean parameter values. We will select 10 locations representing major soybean production areas (NASS, 2023), including 5 environments in Arkansas (Rohwer, Stuttgart, Marianna, Keiser, and Harrisburg) and 5 in Missouri (Portageville, Mount Vernon, Novelty, Garden City, and Albany). The simulation sites cover a geographical gradient of approximately 1000 km. Forty years of historical daily weather data (1985–2024) will be retrieved from NASA/POWER (Stackhouse et al., 2015) and CHIRPS (Funk et al., 2015) platforms, which have proven robust for use in crop modeling (Battisti et al., 2019; Luetkemeier et al., 2018).

Soil inputs will be obtained from the SSURGO database (Soil Survey Staff, 2023), and sowing date information will be derived from USDA-NASS 50% state sowing progress (NASS, 2023). Row spacing will be 76 cm with a plant density of 30 plants m⁻² and planting depth of 35 mm, following typical recommendations for the study regions. We will focus on model outputs beyond final seed yield to enhance fundamental understanding of soybean yield formation. The following model outputs will be analyzed: seed yield, total biomass at flowering and maturity, LAI at flowering and maturity, days to flowering and maturity, and daily crop growth rate. Data analysis will be performed using the R software (R Development Core Team, 2021). Our focus will be on soybean production systems in the U.S. Mid-South, but the approaches implemented here could serve as a model for similar research in other production systems and environments.

Expected results from objective 3. Our benchmark analysis will provide an additional, independent testing of the new genomic prediction-based crop model against well-established genomic prediction models. The optimization analysis will generate a dataset that includes optimum trait values per environment and additional high temporal resolution model outputs such as crop biomass, LAI, and daily crop growth rate.

Potential Pitfalls and Alternative Strategies. The sensitivity and optimization analysis will require high computational demand, which can be overcome using available Arkansas High-Performance Computing Center resources. The proposed framework may have equal or lower accuracy than the GBLUP model. Yet, it may capture phenotypic variance attributed to physiological and environmental adaptation. An ensemble of diverse models may result in superior predicted values, rather than relying simply on genomic information (Tomura et al., 2025), and will be explored in the present proposal. While our preliminary results indicate that longer seed fill provides a yield advantage in the US Mid-South, our hypothesis may not be confirmed in environments with unfavorable environmental conditions during the reproductive period that may preclude adequate seed growth. If this were to occur, we would explore an additional hypothesis that seed yield is modulated by total biomass accumulation (Rowntree et al., 2014).

Dissemination of Results

We will disseminate our findings through peer-reviewed papers and scientific conferences. We anticipate 2 peer-reviewed publications from objective #1 (1 per task), 1 publication from objective #2 (establishment of the genomic prediction-based crop model framework), and 2 publications from objective #3 (1 per task). Scientific conferences include the ASA-CSSA-SSSA International Annual Meeting and the National Association for Plant Breeding (NAPB) annual conference. We will communicate the results to stakeholders through the Agricultural Communications Services with the Arkansas and Missouri Agricultural Experiment Stations and the Arkansas soybean breeding program website. PI/Co-PIs will lead the organization of a webinar with the ASA and CSSA communities on the topic “Integrating Genomic Prediction and Crop Growth Models” to disseminate results from this project and leverage collaborative work. PI Elli will disseminate results from this proposal into his graduate-level Crop Physiology course and undergraduate-level Soybean Production course. Co-PI Vieira will incorporate applications of this project into his cross-listed undergraduate-graduate level Soybean Breeding course. Co-PI Fritsch will integrate information from this project into his Crop Physiology and Grain Crops courses.

Project Timetable

Study Objectives and Benchmarks	Year 1				Year 2				Year 3			
	Q1	Q2	Q3	Q4	Q1	Q2	Q3	Q4	Q1	Q2	Q3	Q4
Objective #1: Identify temperature and photoperiod interactions on developmental rates of diverse soybean accessions												
Task 1.1: Phenotype diverse soybean accessions for phenology traits												
Task 1.2: Generate cultivar-specific coefficients in APSIM-Soybean												
Objective #2: Develop a genomic prediction-based crop model framework through the implementation of genomic prediction models into the APSIM-Soybean model												
Task 2.1: Develop genomic prediction models for relevant cultivar-specific phenology coefficients used in APSIM												
Task 2.2: Implementing the genomic prediction models into the APSIM-Soybean model												
Objective #3: Identify soybean ideotypes for target environments using an integrated genomic prediction-based crop modeling framework												
Task 3.1: Establish a benchmark analysis of the genomic prediction-based crop modeling framework												
Task 3.2: Identifying ideotypes for target environments												